

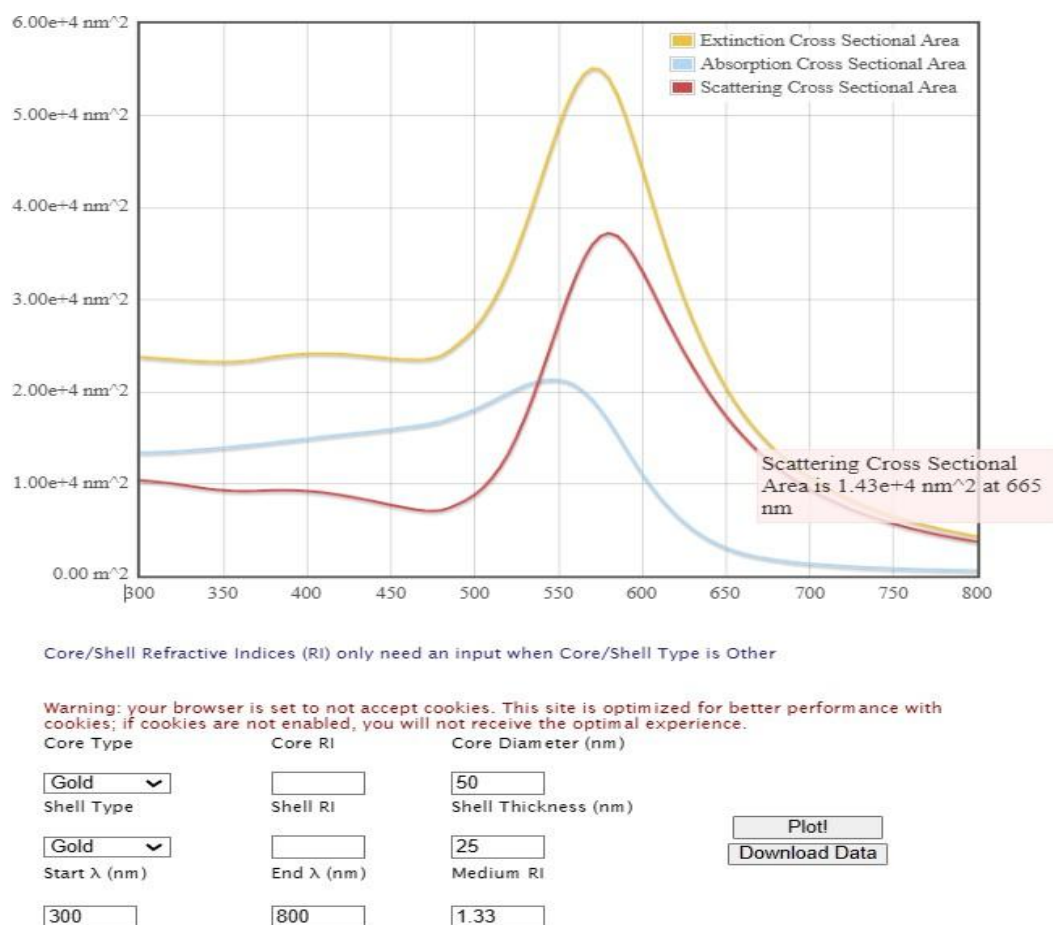
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Exp. 1. Simulation of scattering and absorption of Multi-layer Sphere and Core-Shell Nanoparticle using Mie-Plot tool

Aim: To simulate and analyze the scattering, absorption, and extinction characteristics of multi-layer

spheres and core-shell nanoparticles using the Mie-Plot tool.

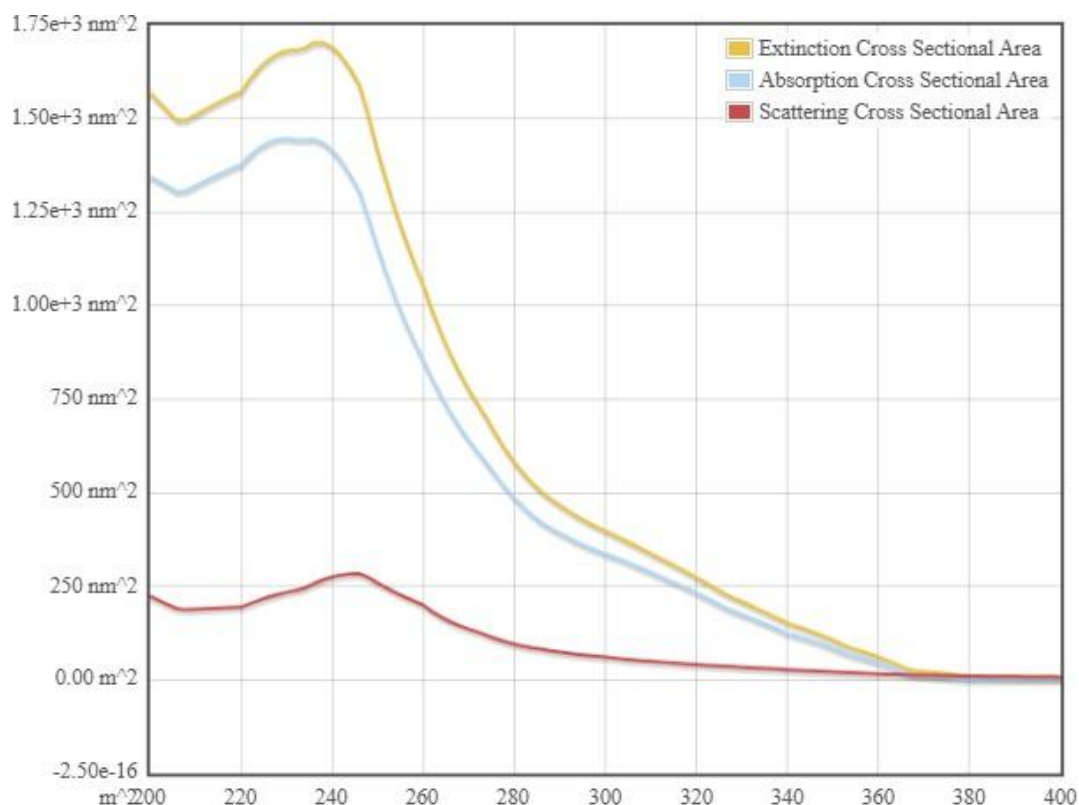
Outcome: Scattering and absorption spectra for a core-shell nanoparticle.



The simulated optical response of the gold core-gold shell nanoparticle shows that the extinction cross-section is the highest among the three optical responses, indicating strong overall interaction with incident light. A prominent resonance peak occurs around 575–590 nm, which indicates the plasmonic response of the gold-based core-shell structure. Near this resonance wavelength, the scattering cross-section increases significantly, showing enhanced scattering of light by the nanoparticle. The absorption

cross-section increases up to approximately 550–560 nm and then decreases rapidly as the wavelength increases. The extinction response is related to both absorption and scattering contributions, following the relationship $\text{extinction} = \text{absorption} + \text{scattering}$. Beyond approximately 650 nm, all three cross-sections gradually decrease, indicating that the interaction of the nanoparticle with incident light becomes weaker at longer wavelengths. Overall, the graph demonstrates that the gold core–gold shell structure has strong plasmonic and optical properties, particularly in the visible wavelength region

Case Study 1: Scattering and absorption spectra for a single-layer sphere (core only).



Core/Shell Refractive Indices (RI) only need an input when Core/Shell Type is Other

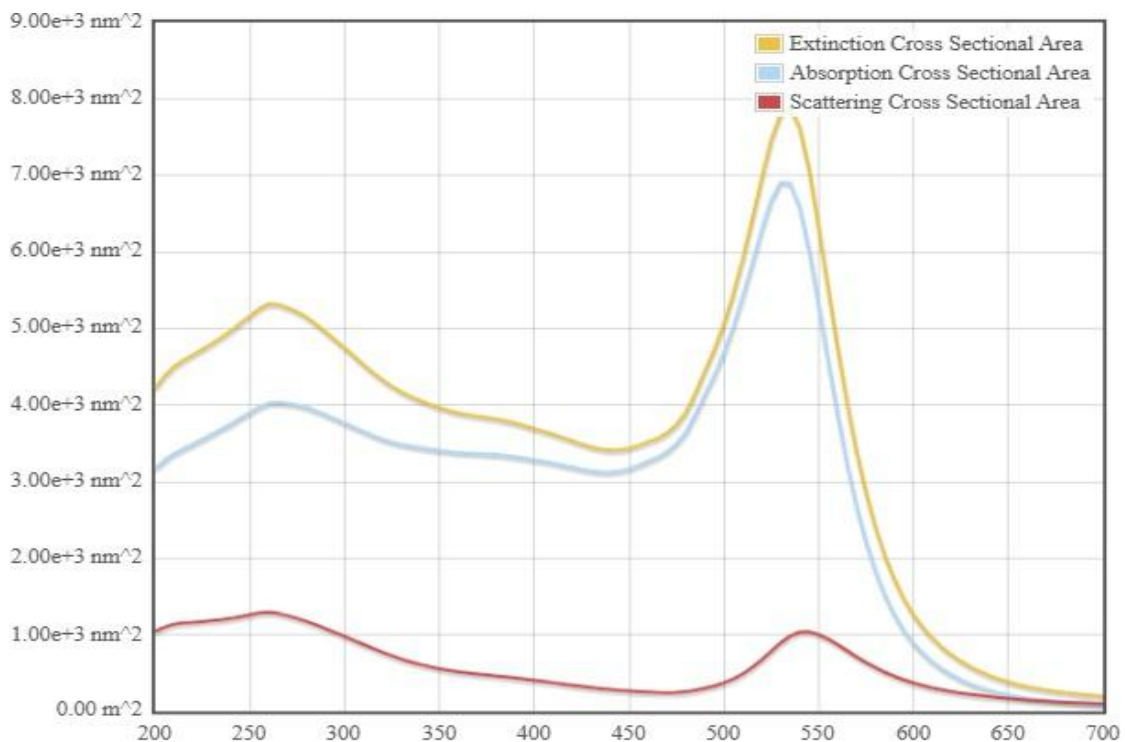
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Core Type	Core RI	Core Diameter (nm)	
TiO<sub>2</sub>		30	
Shell Type	Shell RI	Shell Thickness (nm)	
None		0	
Start λ (nm)	End λ (nm)	Medium RI	
200	400	1.33	

The simulated optical response of the TiO₂ nanoparticle shows that the extinction cross-section reaches its maximum in the 235–245 nm wavelength range, which lies in the ultraviolet (UV) region. This indicates that the TiO₂ nanoparticle interacts strongly with UV light. The absorption cross-section is higher than the scattering cross-section over most of the simulated wavelength range, showing that absorption is the dominant light-interaction mechanism. The scattering remains relatively low because the

particle size is small, resulting in limited scattering of incident light. Beyond approximately 250 nm, the extinction, absorption, and scattering cross-sections gradually decrease as the interaction between TiO_2 and light becomes weaker. Unlike metallic nanoparticles such as gold and silver, TiO_2 does not show a strong visible-region localized surface plasmon resonance (LSPR) because it is a dielectric/semiconductor material rather than a plasmonic metal. Therefore, the graph indicates that the 30 nm TiO_2 nanoparticle mainly exhibits UV absorption with comparatively weak scattering, making its optical response different from that of gold nanoparticles.

Case Study 2: Simulate the optical response of a gold (Au) nanoparticle of 50 nm diameter in water medium.



Core/Shell Refractive Indices (RI) only need an input when Core/Shell Type is Other

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Core Type	Core RI	Core Diameter (nm)
Gold		50
Shell Type	Shell RI	Shell Thickness (nm)
None		0
Start λ (nm)	End λ (nm)	Medium RI
200	700	1.33

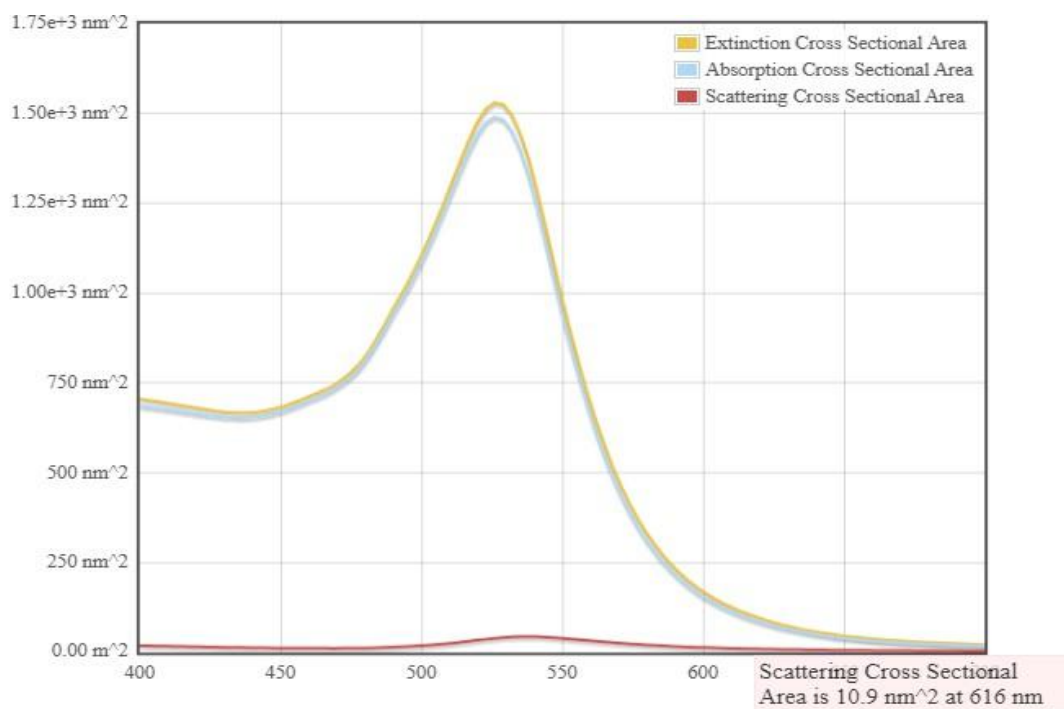
Plot!
Download Data

The simulated optical response of the 50 nm gold nanoparticle shows a strong localized surface plasmon resonance (LSPR) peak around 535–545 nm, which lies in the visible region of the electromagnetic spectrum. At this resonance wavelength, the extinction cross-section reaches its maximum, indicating strong interaction between the incident light and the gold nanoparticle. The absorption cross-section is significantly higher than the scattering cross-section, showing that absorption is the dominant optical process for

this nanoparticle size. The scattering contribution remains relatively small across most of the wavelength range. After approximately 560 nm, the extinction, absorption, and scattering cross-sections decrease rapidly, indicating a reduction in the nanoparticle's interaction with light. Overall, the graph demonstrates that gold nanoparticles have strong optical absorption and plasmonic activity in the visible region, making them useful for applications such as optical sensing, biosensing, imaging, and photothermal applications.

Case Study 3: Simulate a core-shell nanoparticle with a gold core (30 nm) and silica shell

(10 nm).



Core/Shell Refractive Indices (RI) only need an input when Core/Shell Type is Other

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Core Type	Core RI	Core Diameter (nm)
Gold		30
Shell Type	Shell RI	Shell Thickness (nm)
Silica		0
Start λ (nm)	End λ (nm)	Medium RI
400	700	1.33

The simulated optical response of the gold core–silica shell nanoparticle shows a clear plasmon resonance peak around 530–540 nm, indicating strong interaction with visible light. The extinction and absorption curves almost overlap throughout the wavelength range, which shows that absorption is the major contribution to the total extinction. The scattering cross-section remains very small, mainly because the silica shell is transparent and does not significantly absorb visible light. The presence of the silica shell slightly changes the position and shape of the plasmon resonance peak compared with the bare gold nanoparticle, demonstrating that the surrounding shell can influence the optical properties. Beyond approximately 550 nm, the extinction, absorption, and scattering values decrease rapidly, indicating weaker light interaction at longer wavelengths. Overall, the silica shell modifies the optical response of the gold nanoparticle while providing better structural stability without introducing significant additional absorption, making gold–silica core–shell nanoparticles useful for optical sensing, imaging, and other nanophotonic applications.

Comparison

The simulation shows that multi-layer sphere and core-shell nanoparticles have different optical responses due to their structure and material composition. Multi-layer spheres generally exhibit more complex absorption and scattering behavior because they contain multiple interfaces through which light interacts, while core-shell nanoparticles have a simpler structure with one core and one shell. Core-shell nanoparticles show more controlled and tunable resonance characteristics by changing the shell material or thickness, whereas multi-layer spheres provide greater flexibility through the use of several different layers. The scattering response can also increase with the overall particle size and number of layers. In both cases, the particle size, material, layer thickness, and surrounding refractive index strongly influence the extinction, absorption, and scattering spectra.

Conclusion

The Mie-Plot simulation successfully demonstrated the optical behavior of multi-layer sphere and core-shell nanoparticles by analyzing their extinction, absorption, and scattering characteristics at different wavelengths. The results showed that the optical response is strongly influenced by the particle size, material composition, layer or shell thickness, and surrounding medium. Core-shell nanoparticles exhibited controllable plasmon resonance and optical properties, while multi-layer spheres showed more complex absorption and scattering behavior due to their multiple interfaces. Overall, the simulation confirms that Mie theory is an effective method for studying light interaction with spherical and core-shell nanostructures and can be useful in designing nanoparticles for applications such as biosensing, photonic devices, medical imaging, and solar-energy applications.